COMPOSITIONS OF PROPER AND OF UNIVERSALLY CLOSED MORPHISMS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Separatedness and finite typeness are preserved by composition. Schemes, Lemma \ref{schemes-lemma-separated-permanence} and Lemma \ref{lemma-composition-finite-type}, cited at proof lines 9912-9914. Give the remaining parts of properness.

Source reference. Original lemma 01W3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2468–2468. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01KU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2063–2063. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01T3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2360–2360. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *The composition of proper morphisms is proper. The same is true for universally closed morphisms.*

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. A composition of closed morphisms is closed. If $X \to Y \to Z$ are universally closed morphisms and $Z' \to Z$ is any morphism, then we see that $Z' \times_Z X = (Z' \times_Z Y) \times_Y X \to Z' \times_Z Y$ is closed and $Z' \times_Z Y \to Z'$ is closed. Hence the result for universally closed morphisms. We have seen that "separated" and "finite type" are preserved under compositions (Schemes, Lemma 01KU and Lemma 01T3). Hence the result for proper morphisms. □